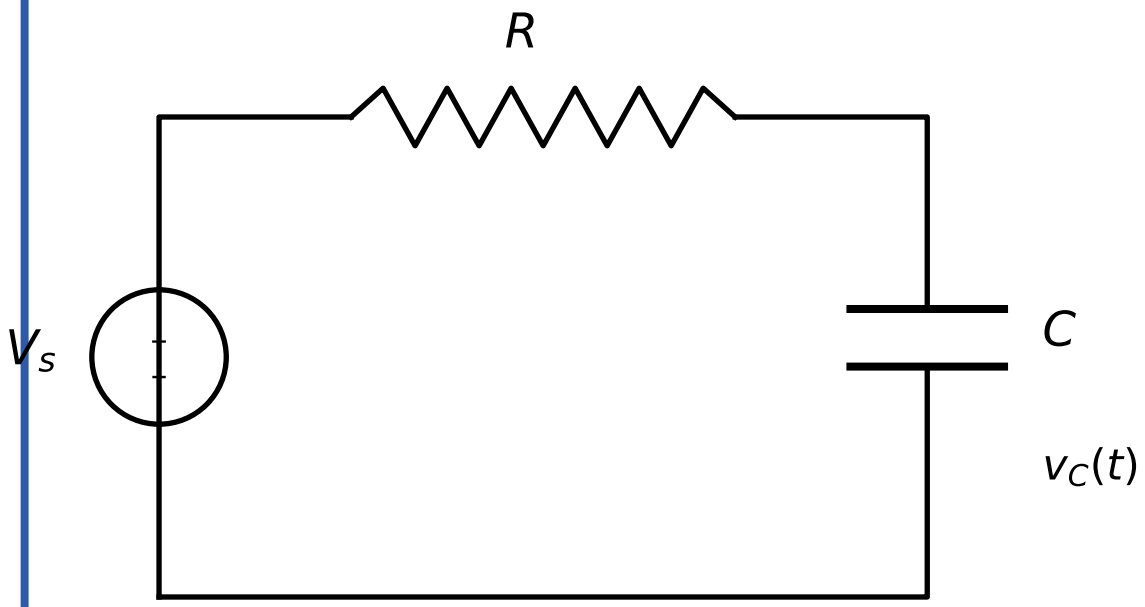
Lecture 5: First-order RC circuits

- Objectives:
  - Write the differential equation of a series RC circuit
  - Find the time constant and the step response
  - Sketch charging and discharging curves
- Reading: course notes, chapter 7 (example only)

Series RC circuit



KVL around the loop:

$$V_s = Ri + v_C, \quad i = C\frac{dv_C}{dt}$$

$$RC\frac{dv_C}{dt} + v_C = V_s$$

first-order ODE!

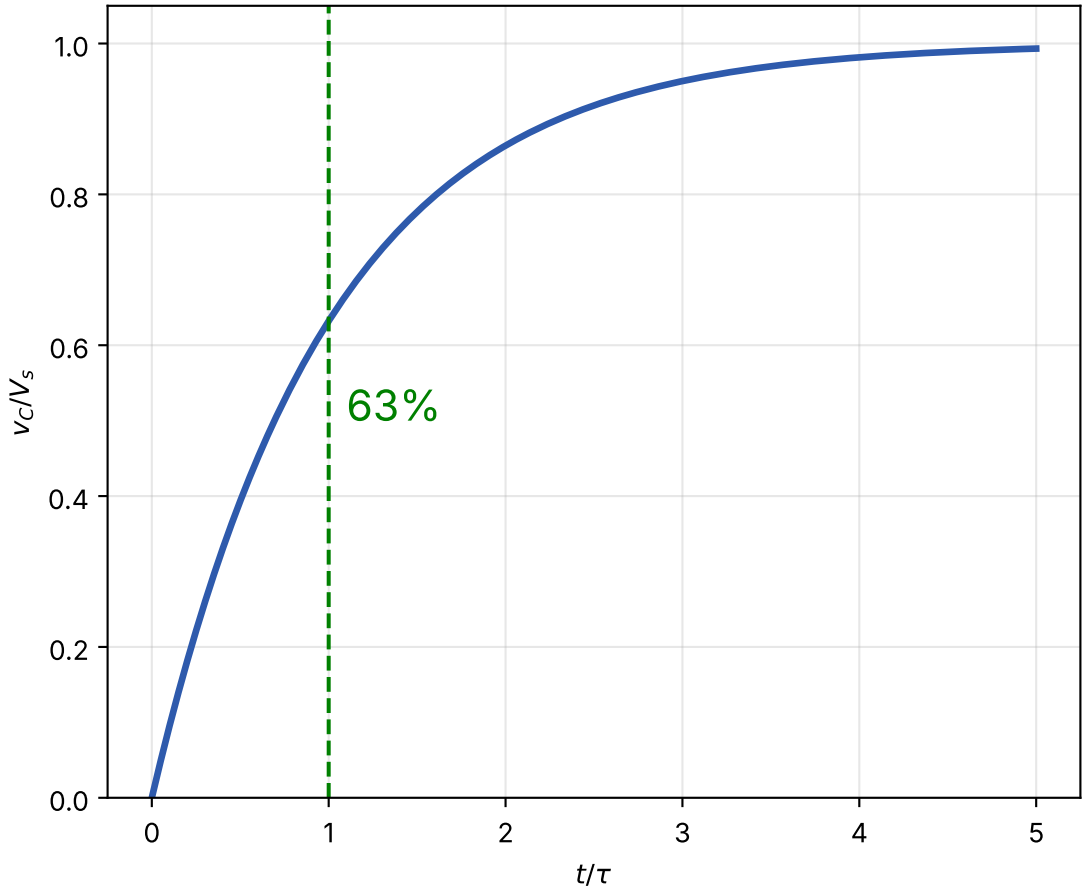
Time constant and step response

$$\tau = RC$$

$$v_C(t) = V_s(1 - e^{-t/\tau}),\ \ t \geq 0$$

Assumes the capacitor starts uncharged.

What fraction of V_s is reached at $t = \tau$?



Discharging

Source removed at $t = 0$, capacitor starts at V_0:

$$v_C(t) = V_0\, e^{-t/\tau}$$

- After 5τ the capacitor is considered discharged (<1%)
- Energy stored: $W = \frac{1}{2}CV^2$

Note to notetaker: explain why 5 tau is used, the way the professor did.

RC vs RL: summary

	RC	RL
Time constant	RC	L/R
State variable	v_C	i_L
Cannot jump	capacitor voltage	inductor current

Think-pair-share

- A camera flash charges a capacitor from a battery.
- With your neighbour: why does the flash need a few seconds
- before it can fire again?
- Be ready to explain using τ.